

Chapter 7: Cell Biology & Genetics

Cell Organelles, Membrane Transport, Cellular Respiration, DNA & Mendelian Genetics

7.1 Cell Structure & Membrane Transport

Cells are the fundamental units of life. Eukaryotic cells feature specialized membrane-bound organelles:

- **Nucleus:** Stores genetic material (DNA).
- **Mitochondria:** Site of ATP generation (cellular respiration).
- **Ribosomes & Endoplasmic Reticulum:** Protein synthesis and lipid metabolism.

Transport across the selectively permeable plasma membrane occurs via passive diffusion, facilitated diffusion, and active transport (requiring ATP).

7.2 Cellular Energy: Photosynthesis & Respiration

- **Photosynthesis:** $6\text{CO}_2 + 6\text{H}_2\text{O} + \text{light} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$ (occurs in chloroplasts).
- **Cellular Respiration:** Glycolysis, the Krebs cycle, and oxidative phosphorylation convert glucose into chemical energy (ATP), yielding roughly 36-38 ATP per glucose molecule.

7.3 Molecular Genetics & Heredity

DNA is a double helix composed of nucleotides (adenine, thymine, cytosine, guanine). The Central Dogma of molecular biology states: DNA $\rightarrow$ RNA $\rightarrow$ Protein.

Mendelian genetics follows the Law of Segregation and the Law of Independent Assortment, predicting phenotypic ratios using Punnett squares.